# Md Sifatullah Sheikh

Dhaka, Bangladesh · mdsifatullahsheikh@gmail.com · +880 1538-061165
Google Scholar · ORCID: 0009-0007-2045-8245 · github.com/SifatSwapnil2022 · linkedin.com/in/mdsifatullahsheikh · Portfolio

## RESEARCH INTERESTS

ML systems, computer vision for autonomous systems, trustworthy AI, and human–computer interaction. My work so far asks one question: when should we trust what a vision model tells us, whether it is judging a face as real or naming a medicinal plant?

## EDUCATION

**B.Sc. in Computer Science and Engineering** Jan 2022 – Jan 2026
*East West University, Dhaka, Bangladesh*
CGPA 3.70/4.00 · Major: Data Science and Intelligent Systems · Dean's List
Thesis: *MediLeafNET: Multimodal Medicinal Plant Identification and Knowledge Retrieval*
Supervisor: Dr. Md. Nawab Yousuf Ali

## PUBLICATIONS

Google Scholar: 20 citations.

**[J1]** Md Al-Imran, **Md Sifatullah Sheikh**, Urmi Kirtonia, Nuzath Tabassum Arthi, Shamim Ripon. "DeFaX: A Cross-Attention Fusion Framework for Robust and Explainable Deepfake Detection." *IEEE Access*, vol. 13, pp. 213962–213979, 2025. doi:10.1109/ACCESS.2025.3645769

**[C1]** **Md Sifatullah Sheikh**, Urmi Kirtonia, Nuzath Tabassum Arthi, Md Al-Imran. "AI-Powered Deepfake Detection Using CNN and Vision Transformer Architectures." *6th International Conference on Big Data Analytics and Practices (IBDAP)*, IEEE, Thailand, 2025. Oral presentation. doi:10.1109/IBDAP65587.2025.11145852

**[D1]** "A Medicinal Plant Leaf Image Dataset of 13 Species Collected in Bangladesh." *Mendeley Data*, Version 2. Contributed to image collection and public release. doi:10.17632/9tdc9gbtgb.2

## RESEARCH EXPERIENCE

**Research Assistant, AI-Generated Content Detection** Dec 2024 – Dec 2025
*Department of CSE, East West University · with Md Al-Imran and Dr. Shamim H Ripon*

- Benchmarked four deep learning architectures for detecting GAN- and AI-generated faces on 140,000 images, and found that CNNs and transformers fail on different cases [C1].
- Designed DeFaX, a cross-attention fusion of Swin Transformer (global context) and EfficientNet (local texture); scaled the study to about 250,000 images with cross-dataset tests [J1].
- Ran ablation studies, layer freezing experiments and Grad-CAM analysis to check that the model looks at manipulated regions, not dataset shortcuts.

**Undergraduate Thesis, MediLeafNET** Feb 2025 – Dec 2025
*East West University · code · live demo*

- Built a system that identifies a medicinal plant from a leaf photo and answers questions about its uses, compounds and safety, combining a vision model with a language model grounded in a curated knowledge base.
- Collected 3,289 leaf images of 11 species at the National Botanical Garden, Bangladesh, and released the extended 13-species dataset publicly [D1].
- Compared several deep learning models on an independent benchmark (best: 94.45% accuracy) and used visual explanations to check which leaf regions drive each prediction.

## PROFESSIONAL EXPERIENCE

**Trainee Software Engineer (AI/ML)** Aug 2026 – Present
*Nasir Syntax Solution Limited (Nasir Group), Dhaka*

- Work on depth estimation for Nasir electric bikes and test machine learning models on real-world data.

**IT Intern** Feb 2026 – May 2026
*Banglalink Digital Communications Ltd., Dhaka*

- Tested IoT devices and networks, monitored sensor data, and helped set up and support IoT solutions; maintained 100+ operational records and reports.

## TEACHING AND MENTORING

**Undergraduate Student Mentor** 2024 – 2025
*Department of CSE, East West University*

- Ran voluntary study sessions on data structures, C++ and object-oriented programming for 40+ junior students.

**Instructor** Jan 2023 – Jul 2024
*Fahim's Tutorial Centre, Dhaka*

- Taught 100+ school and college students; prepared and graded tests and gave each student individual feedback.

## SELECTED PROJECTS

**SkinCare AI: Skin Lesion Detection with an LLM Advisor** May 2026
*github.com/SifatSwapnil2022/SkinCareAI*

- End-to-end pipeline from lesion detection (YOLOv8, mAP@0.5 of 95.4%) to an ensemble classifier (ResNet50, EfficientNet-B0, MobileNetV2) to LLM-generated advice, served with FastAPI and Docker.

## HONORS AND AWARDS

Finalist, ICT Olympiad Bangladesh, Season 3 (national competition) 2026
Dean's List and Dean's Scholarship, East West University (three consecutive semesters) 2023 – 2025
Medha Lalon Merit Scholarship, East West University 2024
3rd Place, IT Olympiad, EWU National RoboFest 2024

## SKILLS

| | |
|---|---|
| **Programming** | Python, C++, SQL |
| **Machine learning** | PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV |
| **Methods** | Vision Transformers, CNNs, cross-attention fusion, object detection, explainability (Grad-CAM, LIME), ablation studies |
| **Tools** | Git, Docker, LaTeX, Google Colab, Kaggle, FastAPI, Streamlit, Gradio |
| **Hardware** | Raspberry Pi, Arduino, sensor integration |
| **Languages** | Bengali (native), English |

## SERVICE AND LEADERSHIP

Executive Member, EWU Robotics Club: organised technical events and workshops 2023 – 2025
Volunteer, LMH Foundation: food and clothing drives for families in need 2015 – Present

## REFERENCES

**Dr. Shamim H Ripon**
Professor, Dept. of CSE
East West University
dshr@ewubd.edu
Co-author [J1]

**Dr. Md. Nawab Yousuf Ali**
Professor, Dept. of CSE
East West University
nawab@ewubd.edu
Thesis supervisor

**Md Al-Imran**
Senior Lecturer, Dept. of CSE
East West University
al.imran@ewubd.edu
Research supervisor [J1, C1]